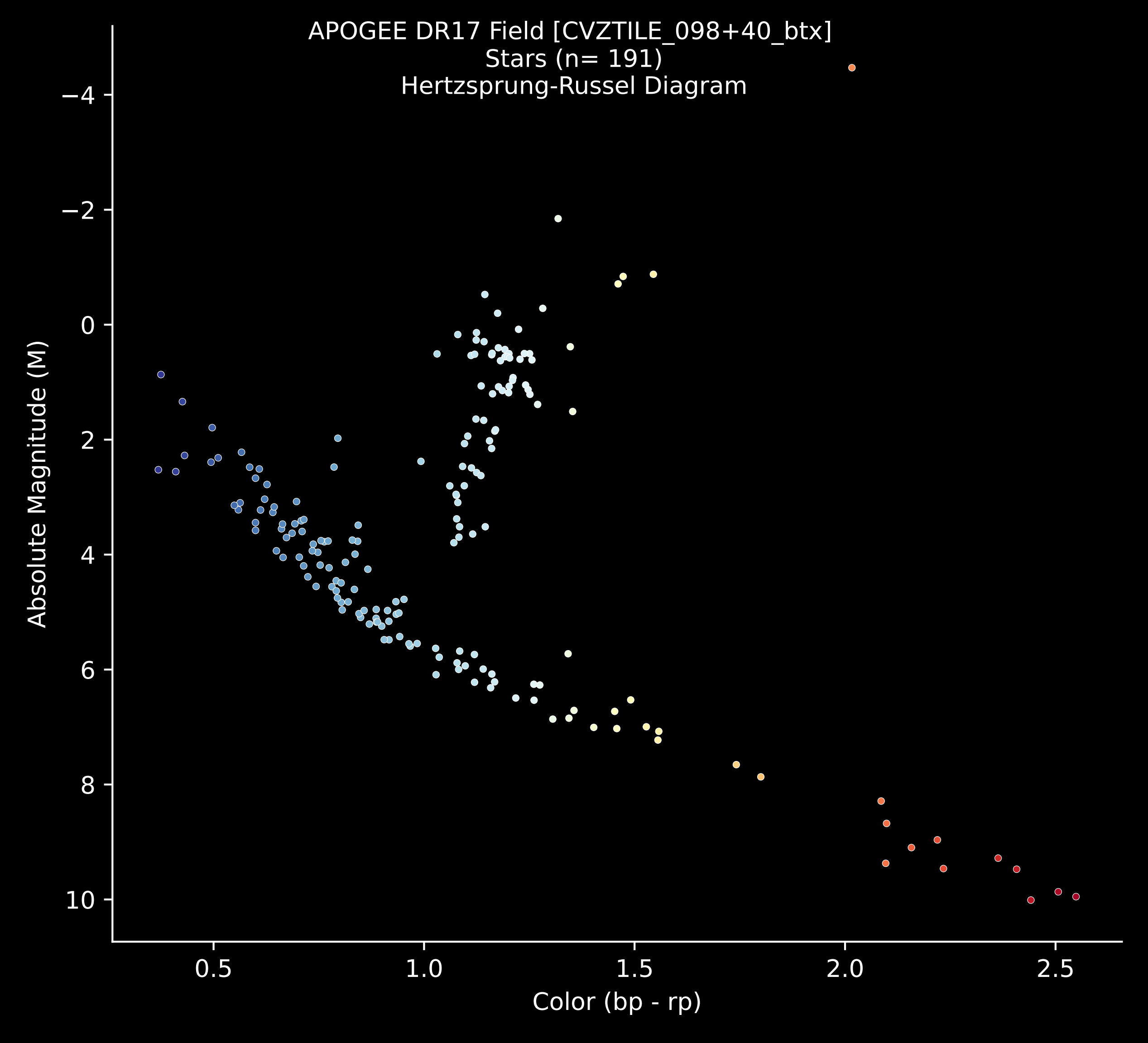
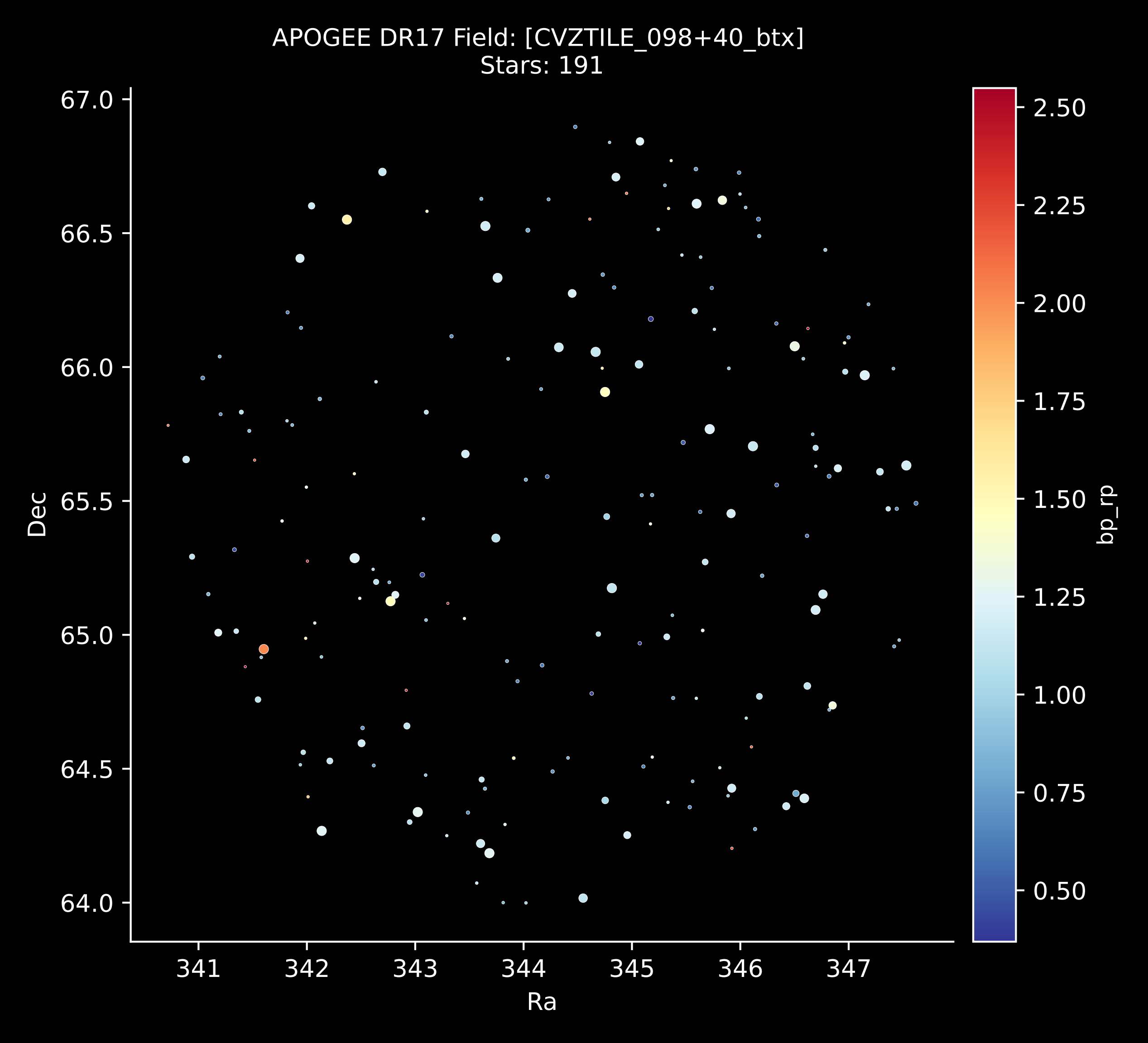
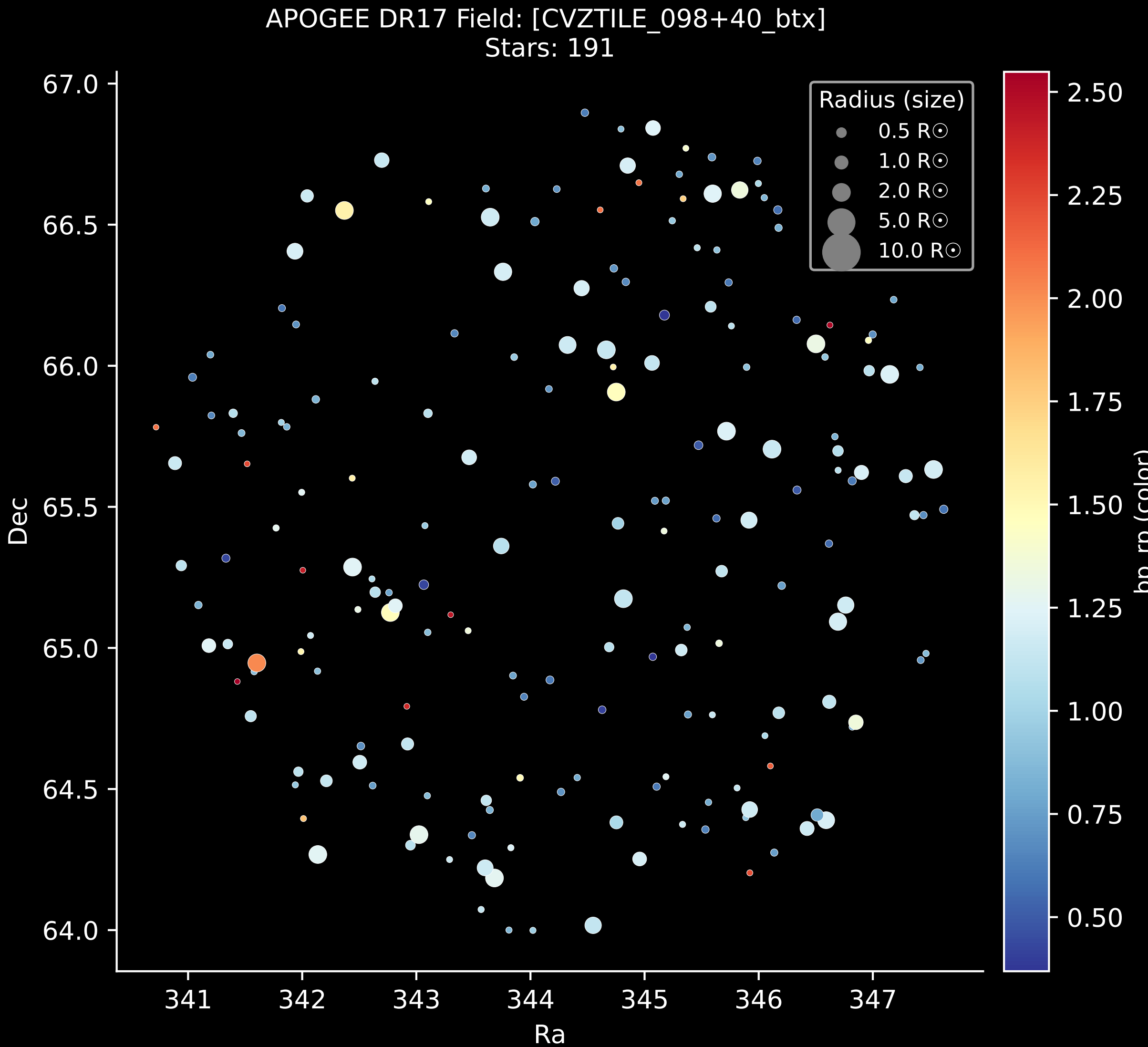
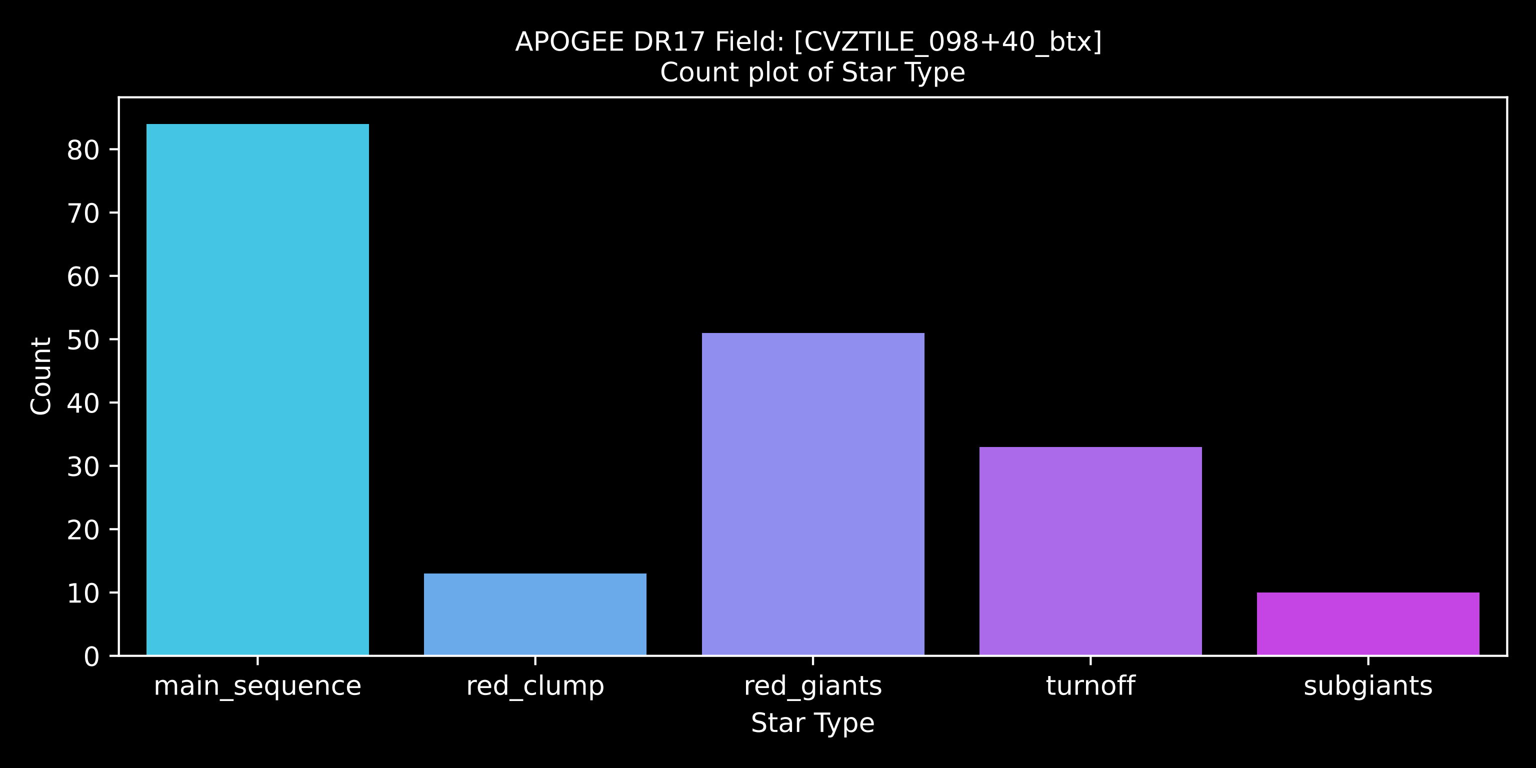
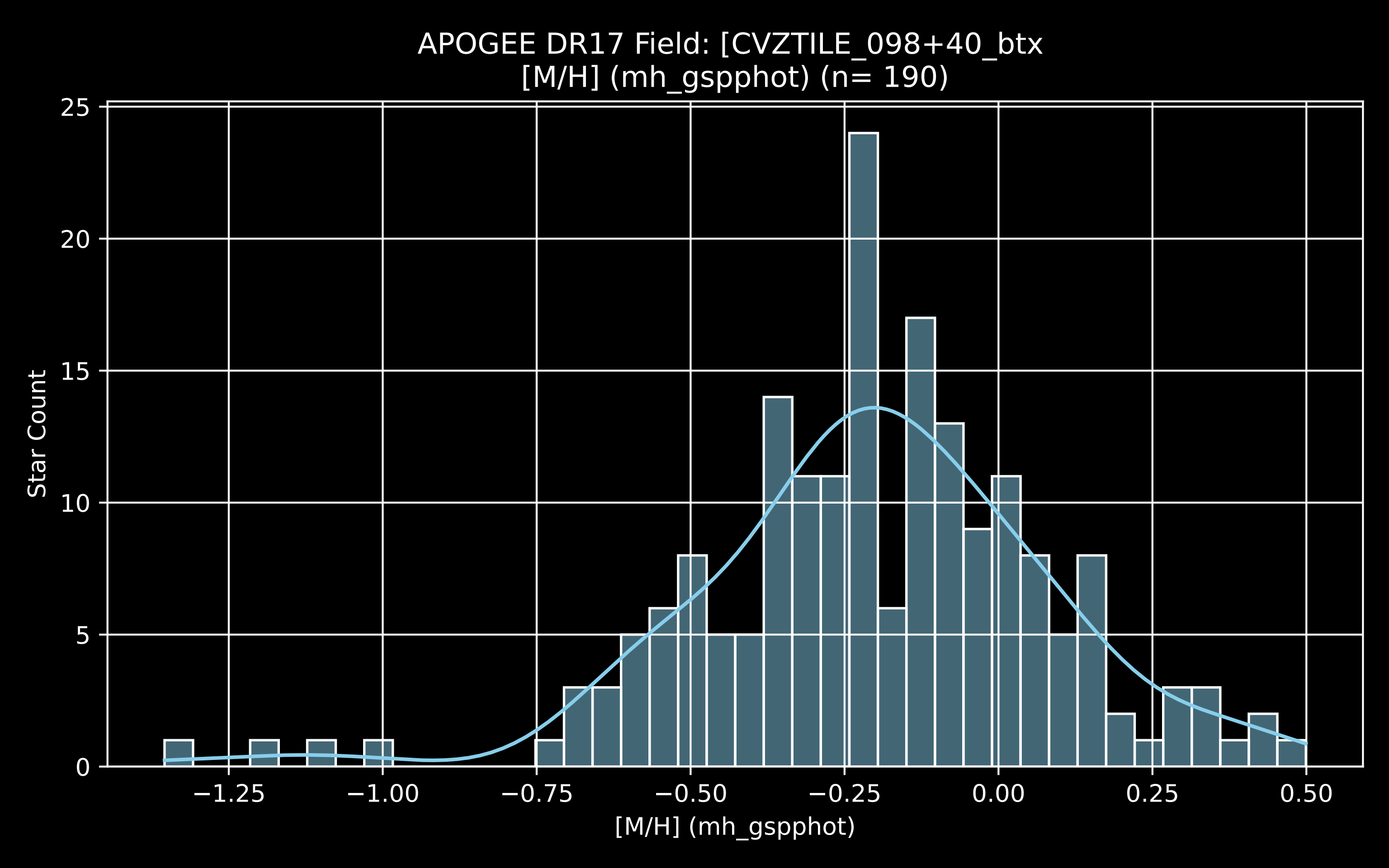

APOGEE DR17 Field [CVZTILE_098+40_btx]
Stars (n= 191)
Hertzsprung-Russel Diagram
Absolute Magnitude (M)
Color (bp - rp)
−4
−2
0
2
4
6
8
10
0.5
1.0
1.5
2.0
2.5

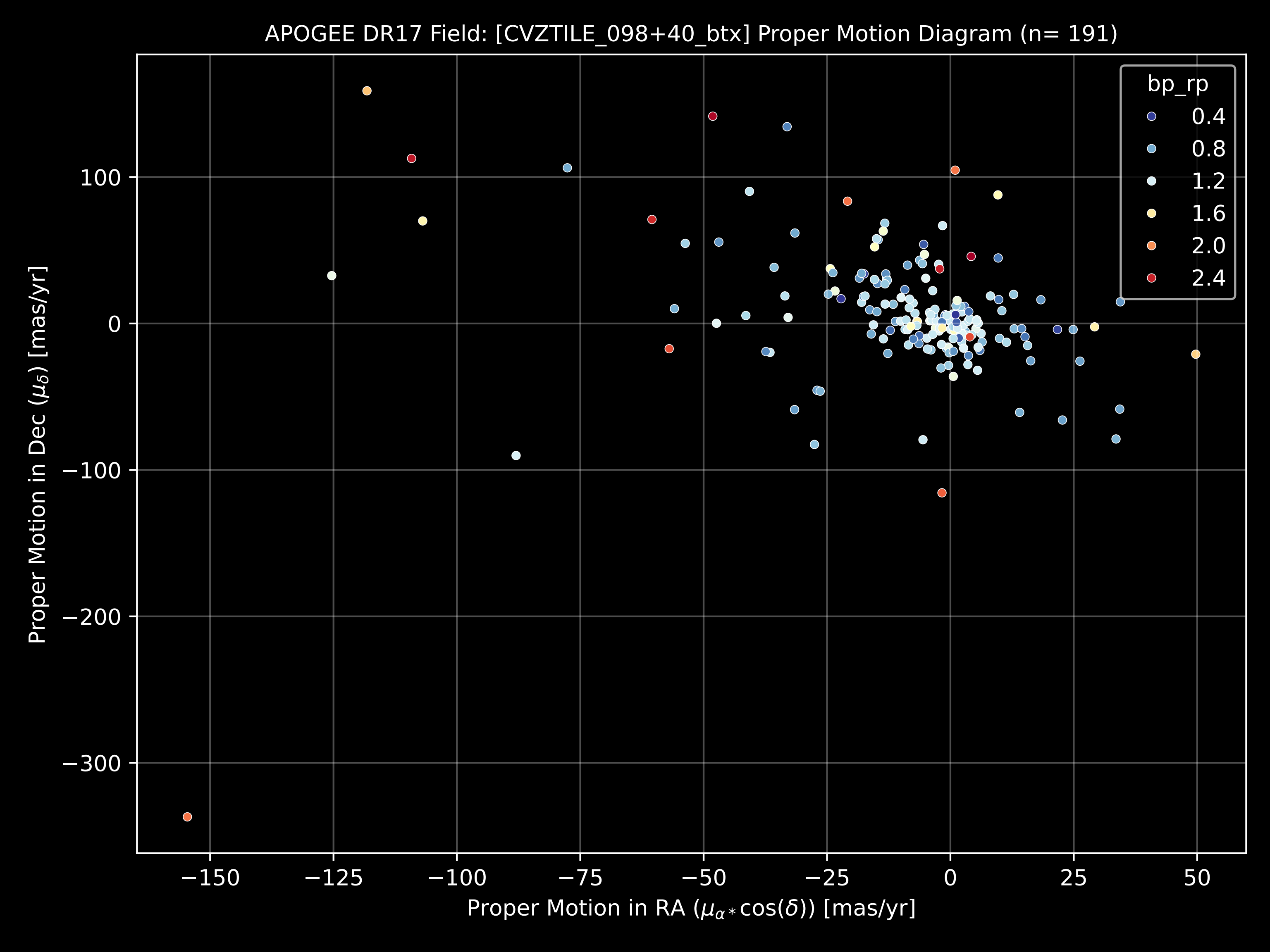

APOGEE DR17 Field: [CVZTILE_098+40_btx]
Stars: 191
Dec
Ra
bp_rp
67.0
66.5
66.0
65.5
65.0
64.5
64.0
341
342
343
344
345
346
347
2.50
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.50

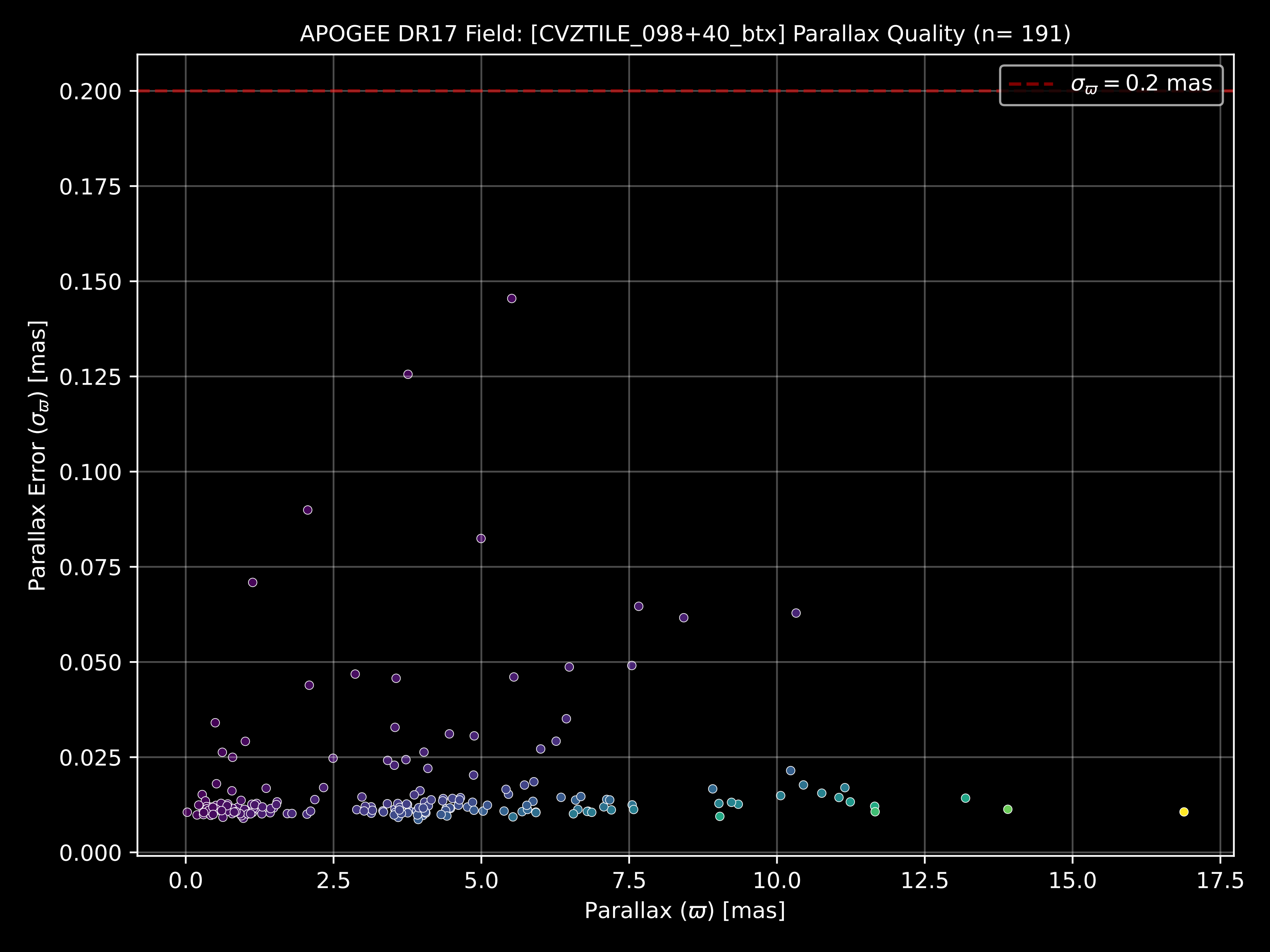


APOGEE DR17 Field: [CVZTILE_098+40_btx]
Stars: 191
Radius (size)
0.5 R☉
1.0 R☉
2.0 R☉
5.0 R☉
10.0 R☉
Dec
Ra
bp_rp (color)
67.0
66.5
66.0
65.5
65.0
64.5
64.0
341
342
343
344
345
346
347
2.50
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.50

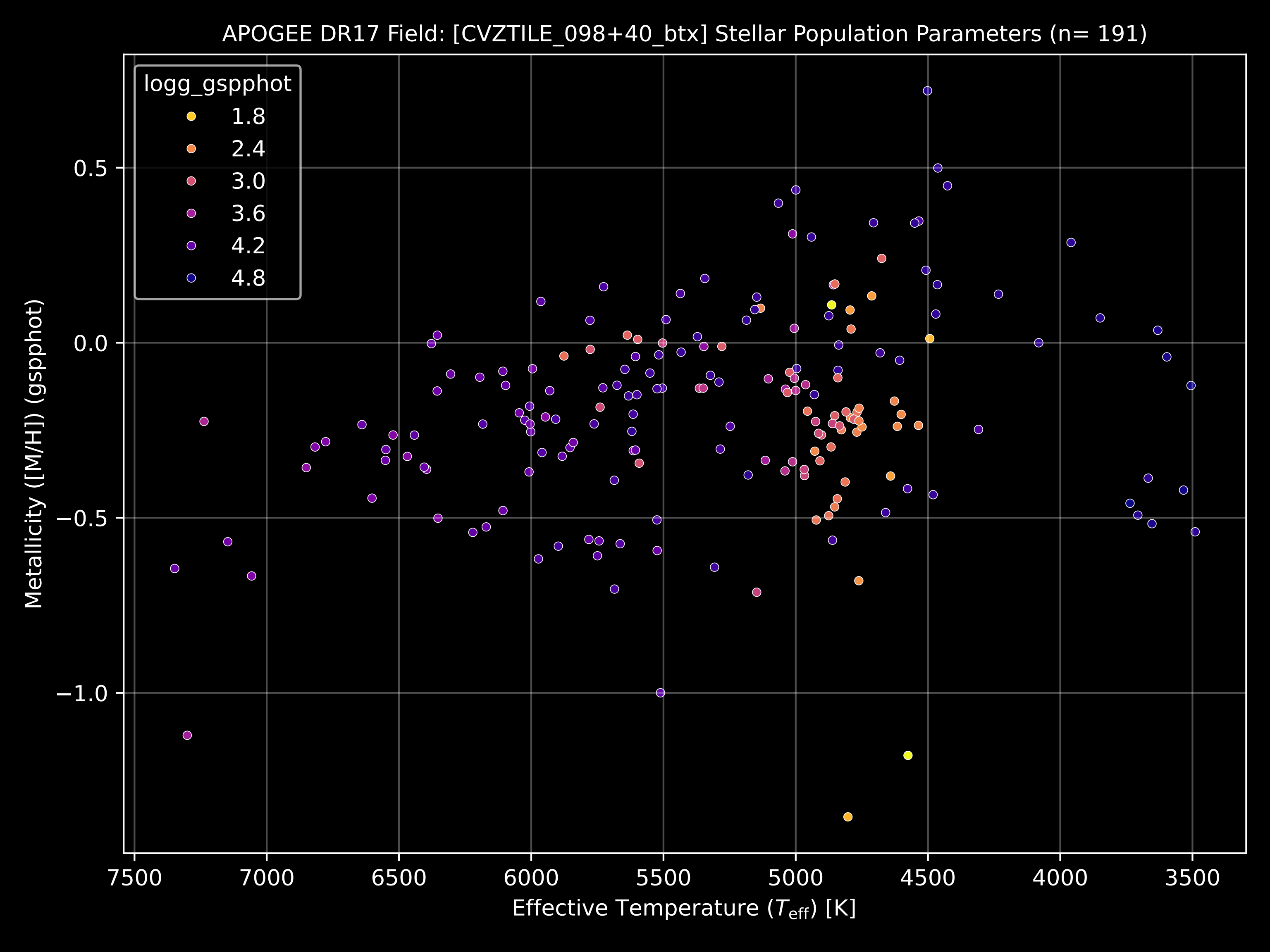

APOGEE DR17 Field: [CVZTILE_098+40_btx]
Count plot of Star Type
Count
0
10
20
30
40
50
60
70
80
main_sequence
red_clump
red_giants
turnoff
subgiants
Star Type

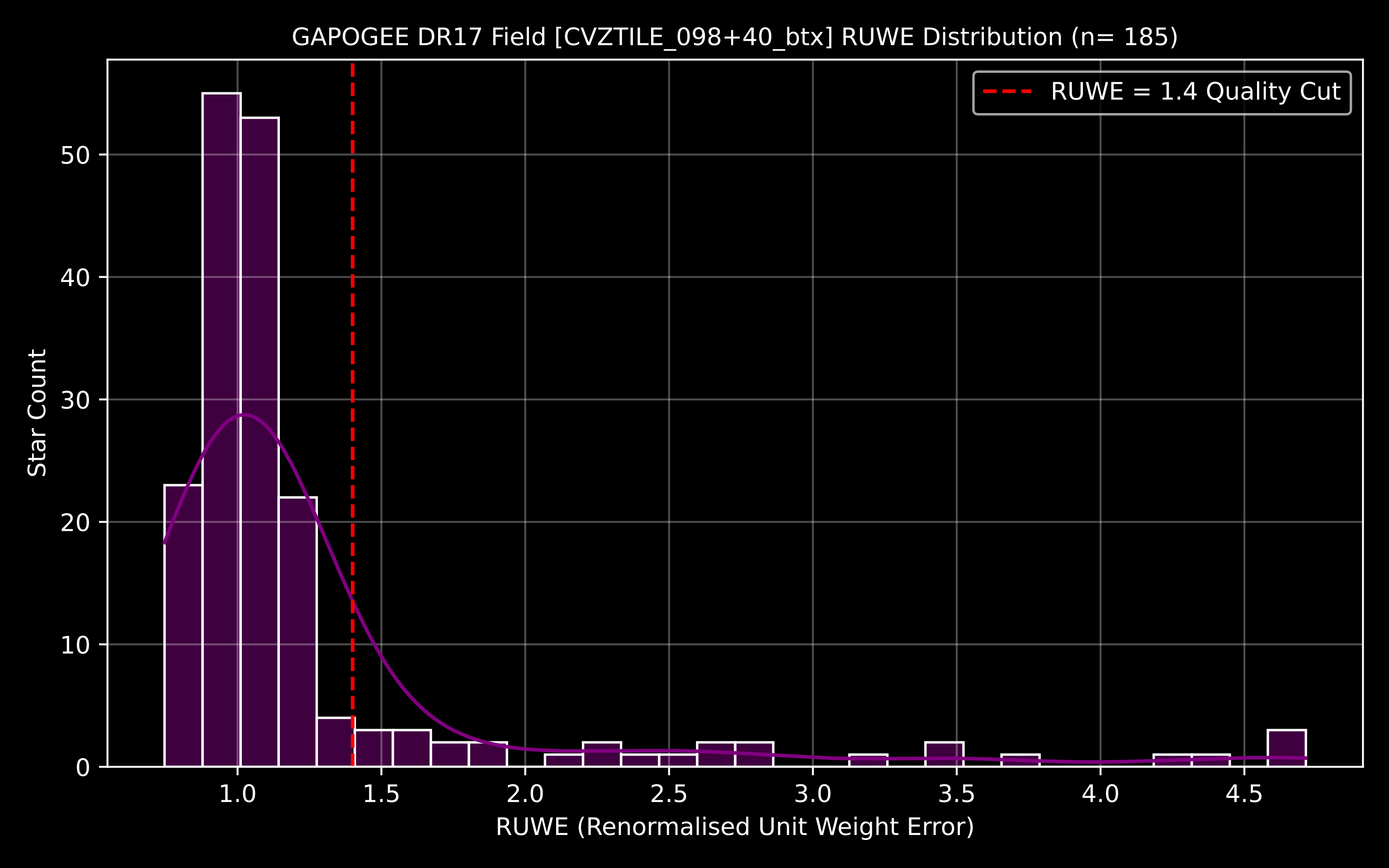

APOGEE DR17 Field: [CVZTILE_098+40_btx
[M/H] (mh_gspphot) (n= 190)
Star Count
[M/H] (mh_gspphot)
0
5
10
15
20
25
−1.25
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50

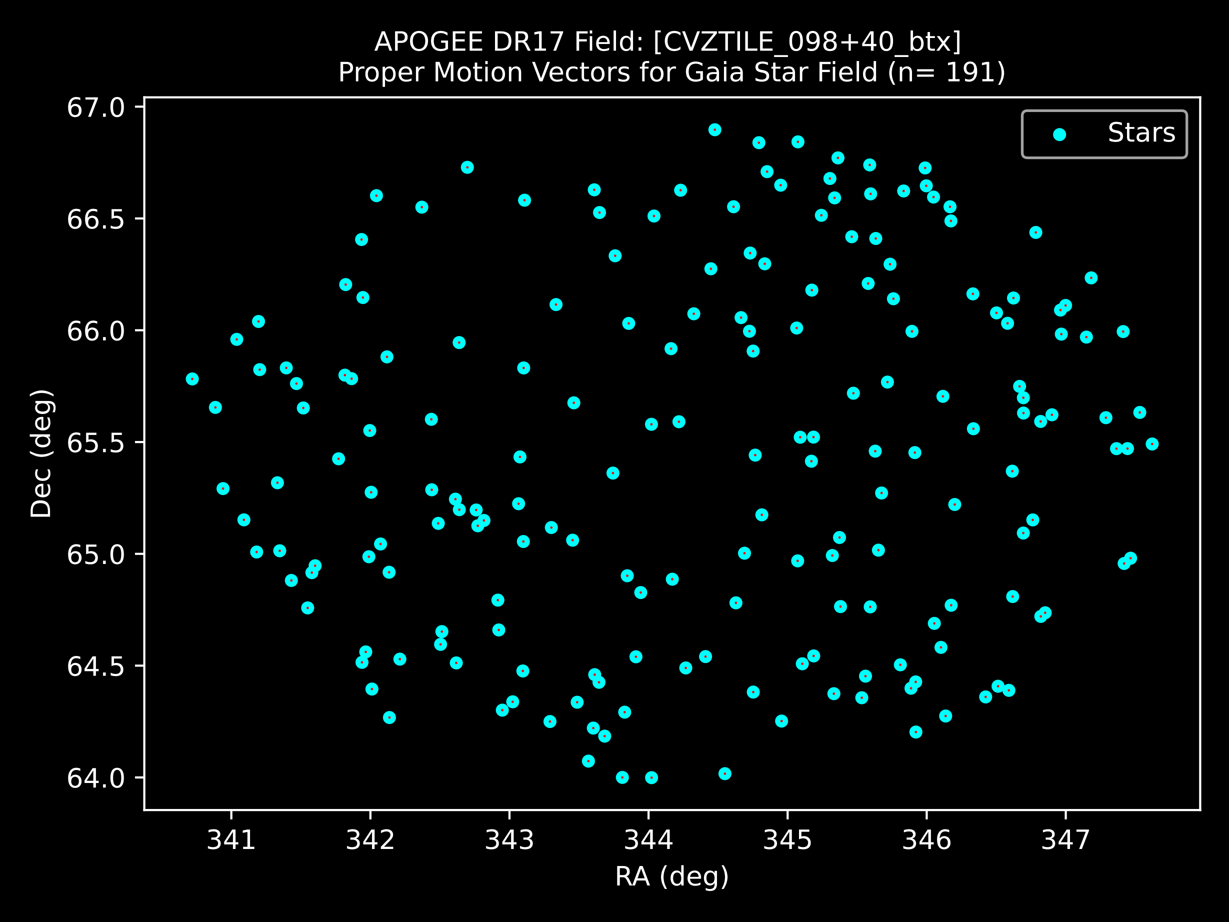

APOGEE DR17 Field: [CVZTILE_098+40_btx] Proper Motion Diagram (n= 191)
bp_rp
0.4
0.8
1.2
1.6
2.0
2.4
100
0
−100
−200
−300
−150
−125
−100
−75
−50
−25
0
25
50
Proper Motion in Dec (μ_δ) [mas/yr]
Proper Motion in RA ($\mu_{\alpha*}\cos(\delta)$) [mas/yr]

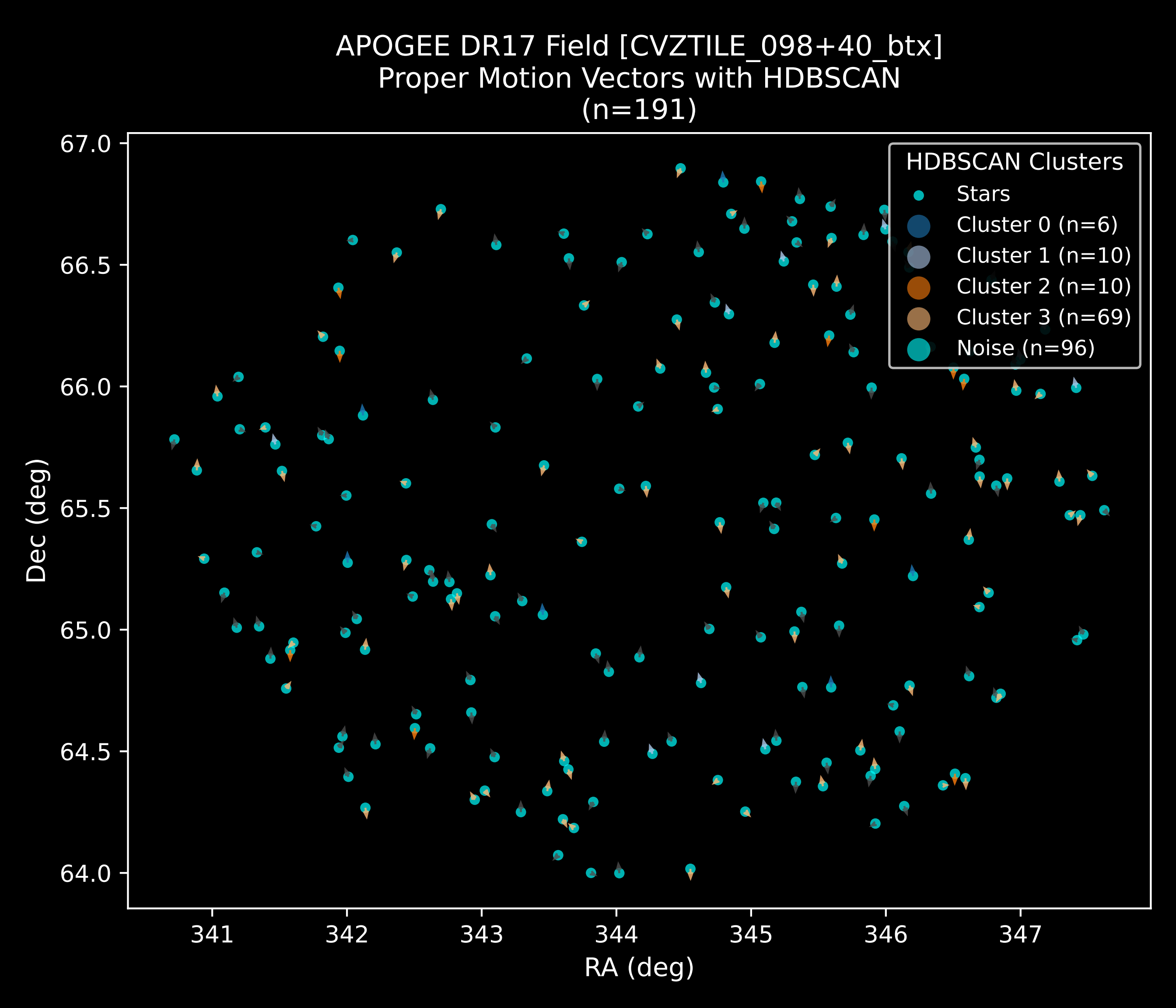


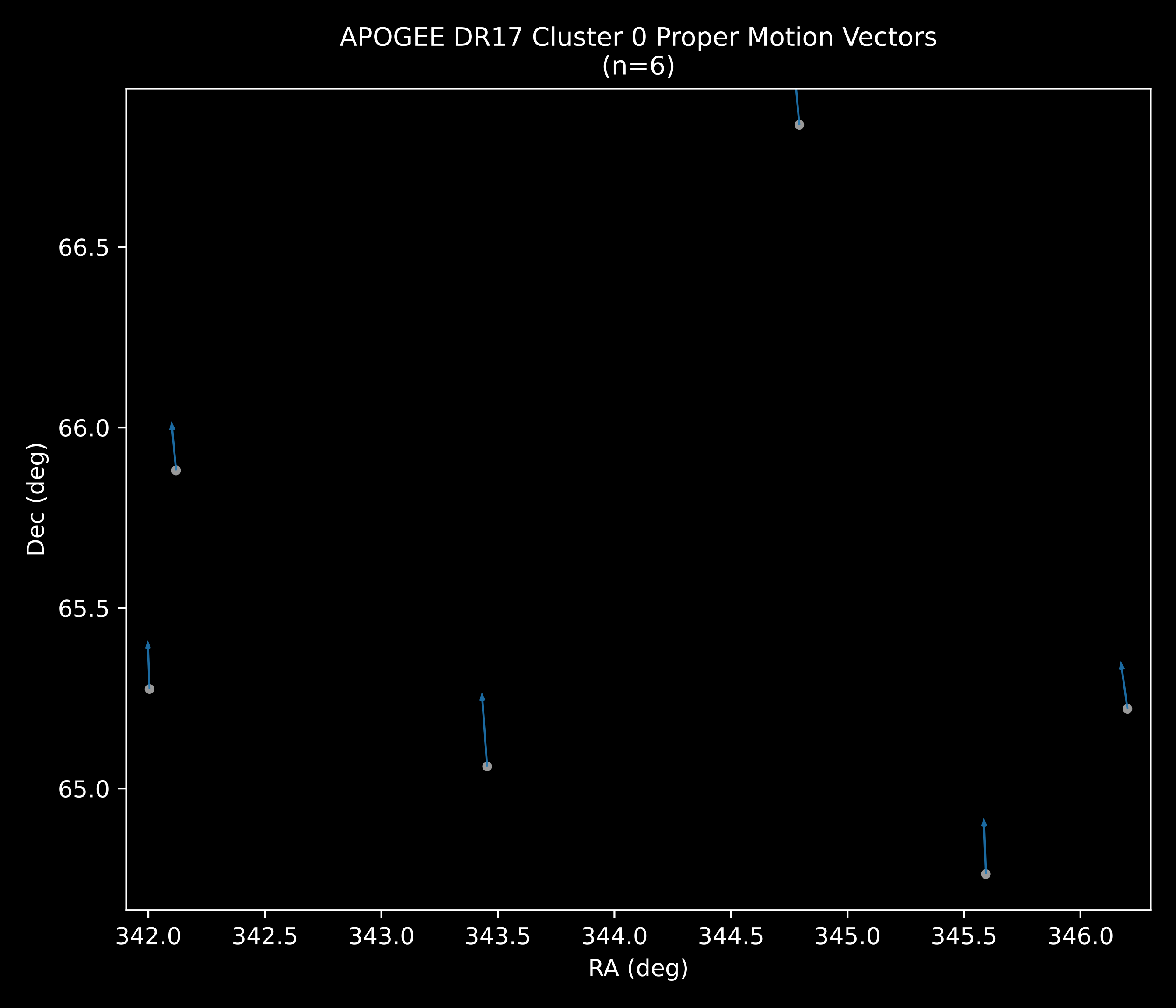
APOGEE DR17 Field: [CVZTILE_098+40_btx] Parallax Quality (n= 191)
$\sigma_\varpi = 0.2$ mas
Parallax Error (σ_ϖ) [mas]
Parallax (ϖ) [mas]
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


APOGEE DR17 Field: [CVZTILE_098+40_btx] Stellar Population Parameters (n= 191)
logg_gspphot
1.8
2.4
3.0
3.6
4.2
4.8
0.5
0.0
−0.5
−1.0
Metallicity ([M/H]) (gspphot)
7500
7000
6500
6000
5500
5000
4500
4000
3500
Effective Temperature ($T_{\rm eff}$) [K]

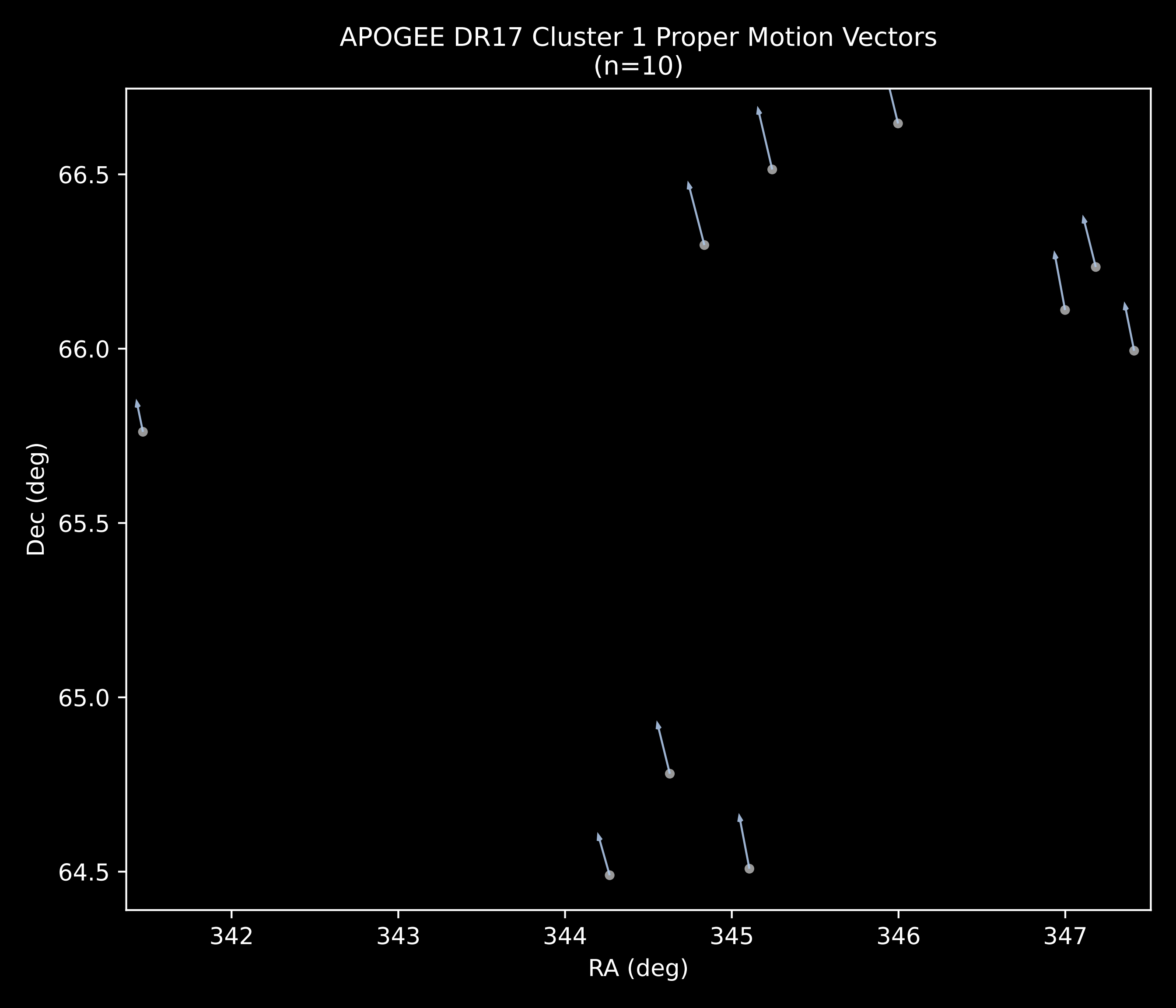

GAPOGEE DR17 Field [CVZTILE_098+40_btx] RUWE Distribution (n= 185)
RUWE = 1.4 Quality Cut
Star Count
RUWE (Renormalised Unit Weight Error)
0
10
20
30
40
50
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5

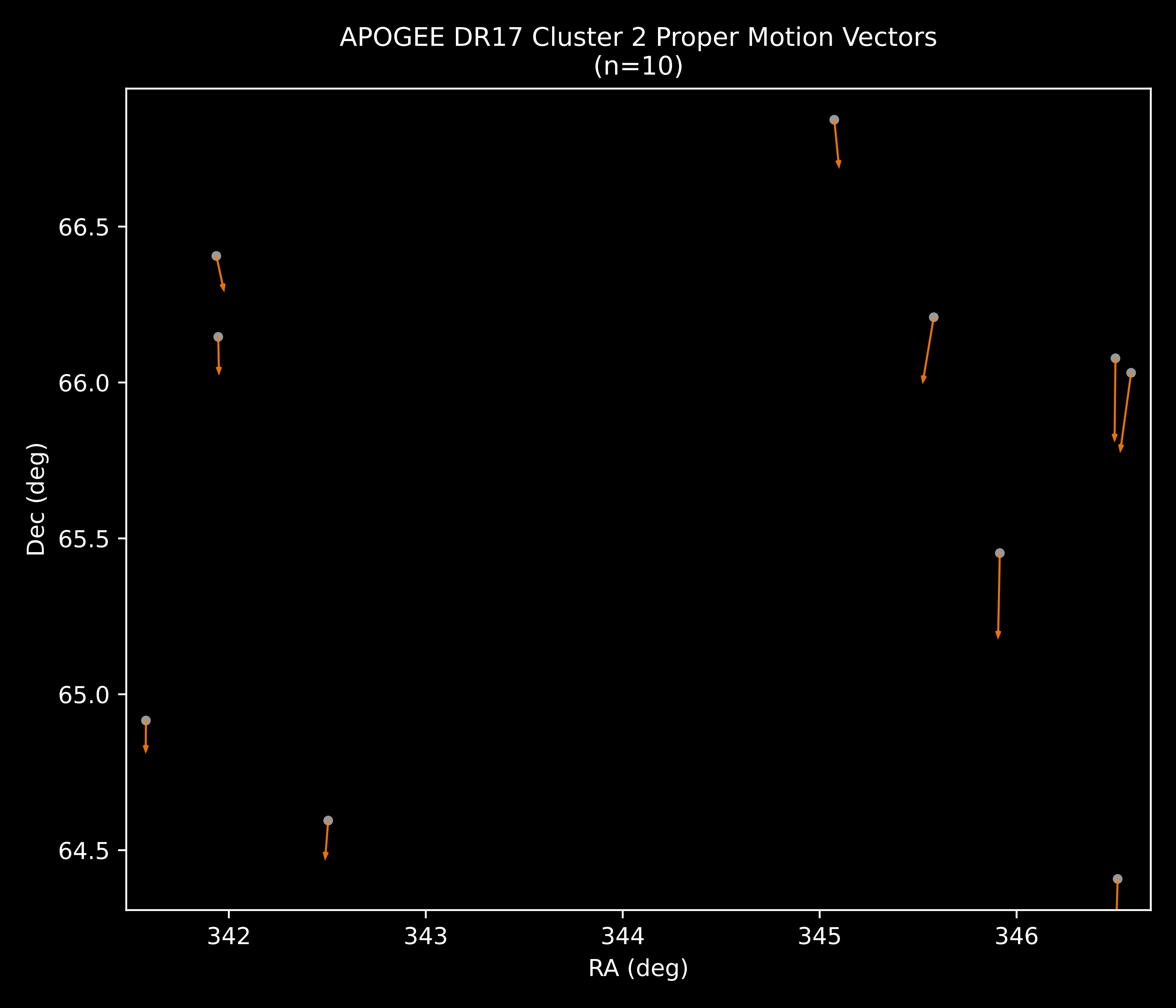

APOGEE DR17 Field: [CVZTILE_098+40_btx]
Proper Motion Vectors for Gaia Star Field (n= 191)
Stars
Dec (deg)
67.0
66.5
66.0
65.5
65.0
64.5
64.0
RA (deg)
341
342
343
344
345
346
347

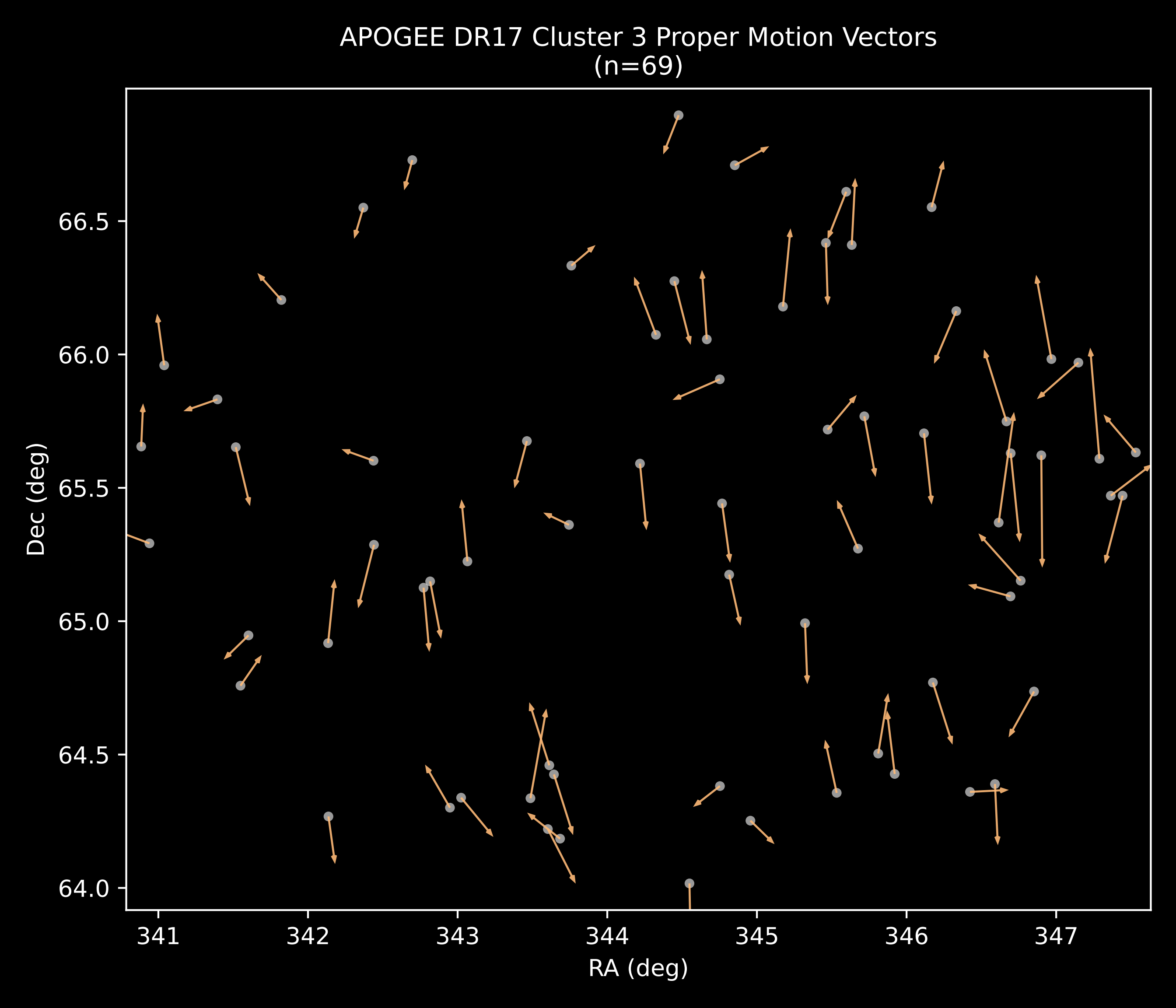

APOGEE DR17 Field [CVZTILE_098+40_btx]
Proper Motion Vectors with HDBSCAN
(n=191)
HDBSCAN Clusters
Stars
Cluster 0 (n=6)
Cluster 1 (n=10)
Cluster 2 (n=10)
Cluster 3 (n=69)
Noise (n=96)
Dec (deg)
RA (deg)
67.0
66.5
66.0
65.5
65.0
64.5
64.0
341
342
343
344
345
346
347



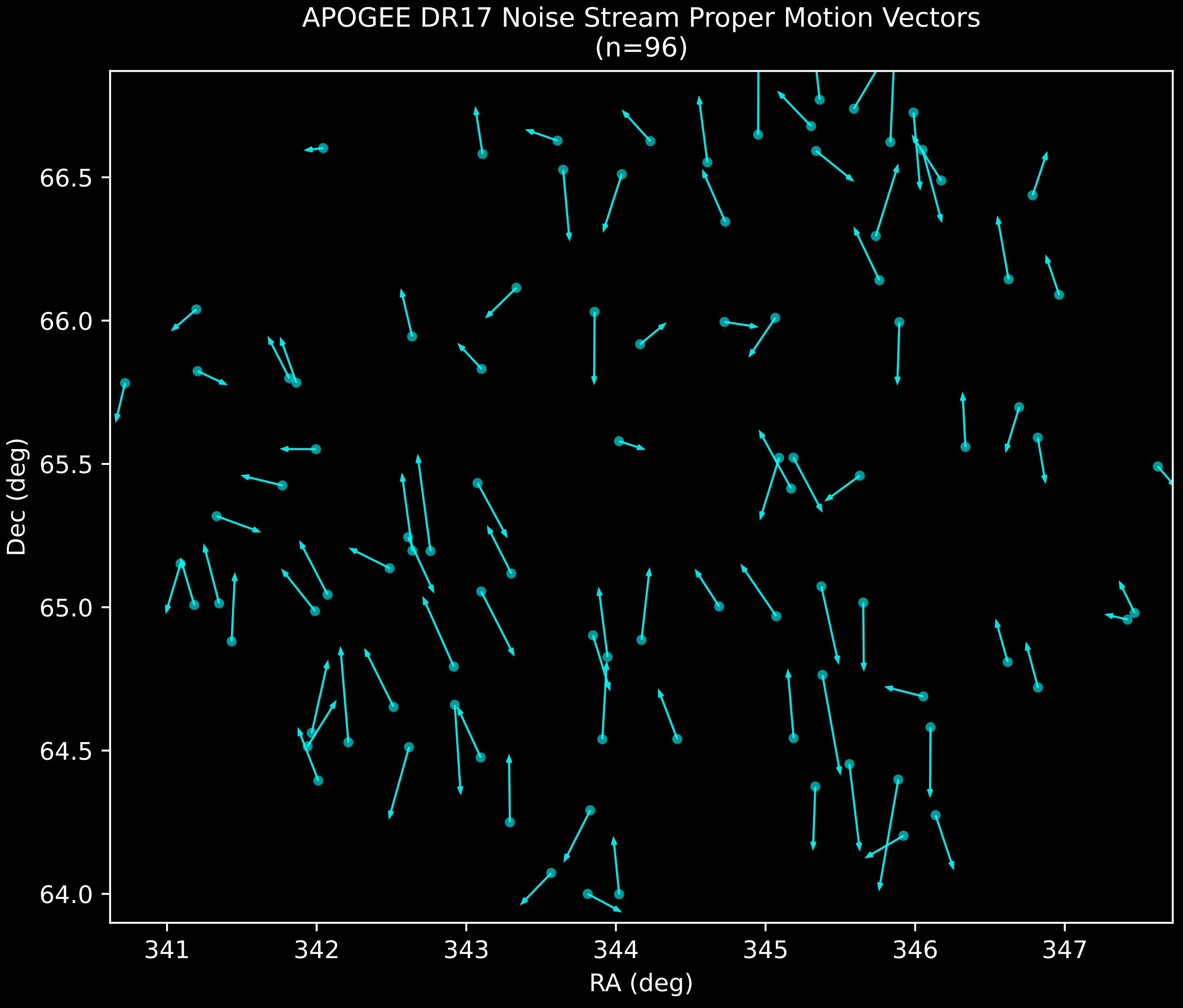
APOGEE DR17 Cluster 0 Proper Motion Vectors
(n=6)
Dec (deg)
66.5
66.0
65.5
65.0
RA (deg)
342.0
342.5
343.0
343.5
344.0
344.5
345.0
345.5
346.0


APOGEE DR17 Cluster 1 Proper Motion Vectors
(n=10)
66.5
66.0
65.5
65.0
64.5
Dec (deg)
342
343
344
345
346
347
RA (deg)


APOGEE DR17 Cluster 2 Proper Motion Vectors
(n=10)
Dec (deg)
RA (deg)
66.5
66.0
65.5
65.0
64.5
342
343
344
345
346



APOGEE DR17 Cluster 3 Proper Motion Vectors
(n=69)
66.5
66.0
65.5
65.0
64.5
64.0
Dec (deg)
341
342
343
344
345
346
347
RA (deg)


APOGEE DR17 Noise Stream Proper Motion Vectors
(n=96)
Dec (deg)
66.5
66.0
65.5
65.0
64.5
64.0
RA (deg)
341
342
343
344
345
346
347